AareML

Predicting River Water Quality in Swiss Catchments

CAS in Advanced Machine Learning
University of Bern — Institute of Computer Science

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Abstract. AareML applies a sequence-to-sequence LSTM, modelled on the LakeBeD-US benchmark (McAfee et al., 2025), to predict dissolved oxygen (DO) and water temperature at 14-day horizons across 115 Swiss river catchments from the CAMELS-CH-Chem dataset (Nascimento et al., 2025). Starting from three statistical baselines, we train a single-site LSTM (Optuna-tuned over 20 trials, best DO RMSE = 0.299 mg/L) and evaluate its zero-shot transfer to all gauges with sufficient DO coverage. SHAP attribution explains which input features and catchment characteristics drive forecast skill. Baseline DO RMSE on gauge 2473 ranges from 0.30–0.34 mg/L, already well below the LakeBeD-US lake reference of 1.40 mg/L, suggesting rivers are more predictable than lakes under the same LSTM architecture. Multi-site and SHAP results will follow in subsequent notebook runs.

Dataset	Method	Targets	Metric	Status
CAMELS-CH-Chem	Seq2Seq LSTM	DO · Temp	RMSE · KGE	In progress

1. Introduction

Dissolved oxygen (DO) is a primary indicator of aquatic health: concentrations below 5 mg/L stress fish populations, while hypoxic events (below 2 mg/L) cause mass mortality. In Swiss river networks, DO dynamics are driven by temperature, photosynthesis, reaeration, and upstream nutrient loads — a complex interplay that statistical models have historically struggled to capture at multi-day forecast horizons.

Recent work on lake water quality has demonstrated that sequence-to-sequence LSTMs can outperform traditional baselines substantially. McAfee et al. (2025) introduced LakeBeD-US, a benchmarking dataset for 139 US lakes with a seq2seq LSTM achieving a test RMSE of 1.40 mg/L for DO at a 14-day forecast horizon. Their architecture — 21-day lookback, teacher-forced decoder, masked MSE loss — provides a reproducible benchmark that we adapt here for river systems.

The CAMELS-CH-Chem dataset (Nascimento et al., 2025), published in early 2025, provides daily high-frequency sensor measurements of temperature, pH, electrical conductivity, and dissolved oxygen at 86 Swiss gauging stations from 1981 to 2020, complemented by 38-variable chemistry grab samples, land cover, and 115 catchment attribute files. To our knowledge, no published machine learning study has yet applied predictive modelling to CAMELS-CH-Chem, making this a genuine first-of-its-kind cross-ecosystem transfer study.

AareML makes four contributions: (1) adapts the LakeBeD-US seq2seq LSTM to Swiss river catchments; (2) establishes three statistical baselines with block-bootstrap confidence intervals; (3) evaluates multi-site generalisation via zero-shot transfer and per-gauge retraining; and (4) applies SHAP attribution to identify which sensor inputs and catchment attributes drive prediction skill.

2. Related Work

2.1 Machine Learning for Water Quality Prediction

LSTMs have become the dominant architecture for hydrological time-series forecasting. Kratzert et al. (2018, 2019) demonstrated that a single LSTM trained across 531 US basins outperforms a calibrated process-based model (HBV) on 43% of catchments, with particular gains in ungauged basins. Their Entity-Aware LSTM (EA-LSTM) incorporates static catchment attributes directly into the gating mechanism, improving cross-basin transfer — a design we may adopt in future work.

For water quality specifically, Zhi et al. (2021) showed that LSTMs predict stream nitrate concentrations substantially better than process-based models at weekly timescales. Barzegar et al. (2020) and others have applied similar approaches to DO prediction in lakes, though primarily at single sites without multi-site generalisation evaluation.

2.2 The LakeBeD-US Benchmark

McAfee et al. (2025) proposed LakeBeD-US as the first standardised benchmark for lake water quality prediction, covering 139 US lakes with daily temperature and DO observations. Their seq2seq LSTM uses a 21-day lookback, 14-day forecast horizon, SAITS imputation for missing

values, and Optuna hyperparameter search across 50 trials. The reported test RMSE of 1.40 mg/L for DO provides our primary reference point. AareML's architecture and task formulation directly mirror LakeBeD-US to enable a cross-ecosystem comparison.

2.3 CAMELS-CH-Chem

The CAMELS-CH-Chem dataset (Nascimento et al., 2025) extends the CAMELS-CH hydrometeorological benchmark with stream water chemistry observations at 86 Swiss gauging stations. The daily sensor files contain temperature, pH, electrical conductivity, and dissolved oxygen from as early as 1981 through 2020. Ancillary data include NAWA FRACHT grab samples (38 variables, 7–14 day intervals) covering nutrients, heavy metals, and discharge proxies, as well as 115 catchment attribute files covering land cover, livestock density, atmospheric deposition, and rain isotopes. No machine learning paper has yet applied predictive modelling to this dataset.

3. Data

3.1 Dataset Overview

Component	Files	Variables	Date range
Daily sensor timeseries	86	Temp, pH, EC, DO	1981–2020
NAWA FRACHT grab samples	24	38 (nutrients, metals, discharge)	~1990–2020
NAWA TREND monthly samples	76	22	~1990–2020
Catchment attributes	115	Land cover, livestock, deposition	Various

Table 1: CAMELS-CH-Chem dataset components.

3.2 Data Availability

Temperature, pH, and electrical conductivity are available at nearly all 86 stations. Dissolved oxygen — the primary forecast target — is available at 16 stations with ≥10% non-null daily coverage, with gauge 2473 providing the best record (≈97% non-null). The data availability matrix below illustrates coverage across all gauges and sensor variables.

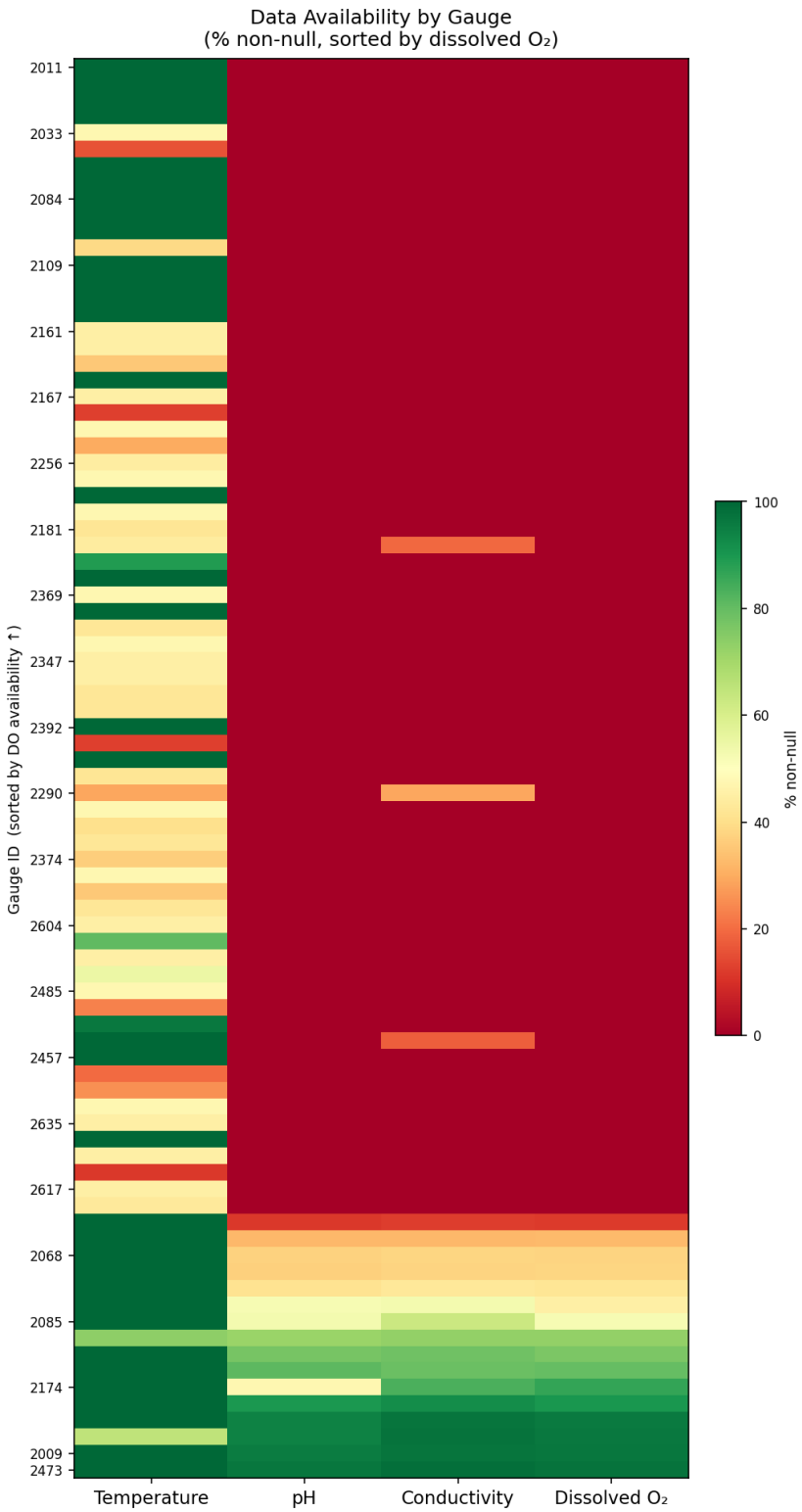

Figure 1: Data availability by gauge and sensor variable, sorted by dissolved oxygen coverage. Green = well-observed; red = sparse.

3.3 Gauge 2473 — Study Site

Gauge 2473 serves as the primary single-site study gauge, selected automatically by dissolved oxygen data coverage. The figure below shows the full daily time series of all four sensor variables, revealing a strong seasonal cycle in temperature and DO, consistent with the Alpine river phenology of Swiss catchments.

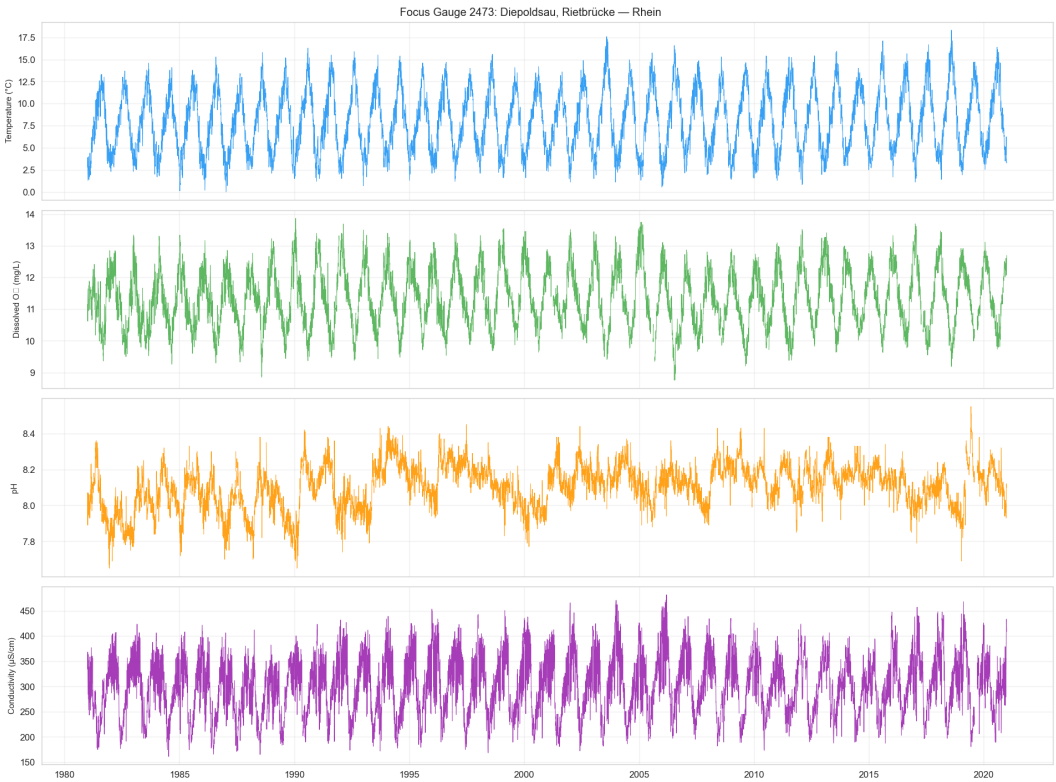

Figure 2: Full daily sensor time series for gauge 2473 (temp, pH, EC, DO). Grey shading marks the test period (2017–2020).

3.4 Seasonality

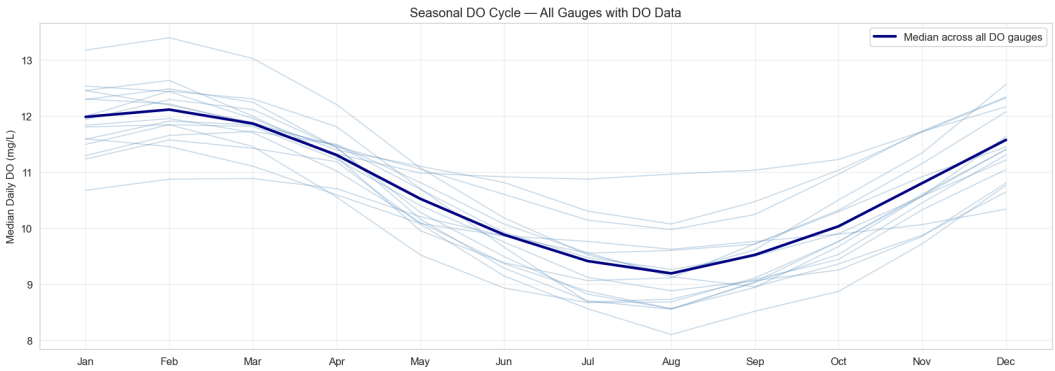

Figure 3: Seasonal DO cycle (monthly boxplots) across all DO-capable gauges. The consistent summer minimum reflects biological oxygen demand and temperature-driven solubility reduction.

3.5 Train / Validation / Test Split

Data are split chronologically to prevent temporal leakage: train ≤ 2014-12-31 (~12,000 daily windows), validation 2015–2016 (~730 days), test 2017–2020 (~1,460 days). All scalers are fitted on training data only and applied identically to validation and test splits. NaN imputation uses per-column training means.

4. Methods

4.1 Task Formulation

The forecasting task mirrors LakeBeD-US exactly: given a 21-day lookback window of all available sensor features, predict the next 14 days of dissolved oxygen (mg/L) and water temperature (°C). Formally, let x_t in R^4 denote the four sensor measurements on day t. Given X = $[x_{t-20}, \ldots, x_t]$, predict Y = $[y_{t+1}, \ldots, y_{t+14}]$ where $y_\tau = (DO_\tau, Temp_\tau)$.

Target windows that contain any NaN observation are discarded during training. Missing values in the input window are imputed with per-column training-set means before feature scaling — a conservative strategy that avoids leaking future information while preserving temporal structure.

4.2 Baseline Models

Three baselines establish the lower bound for model comparison:

- Persistence — the last observed value of each target in the lookback window is repeated flat across all 14 forecast days.
- Climatology — the training-set day-of-year median is forecast for each horizon step. Vectorised lookup; no per-window computation.
- Ridge Regression — a Ridge model trained per target per horizon step (28 models total) on the flattened 21-day input window (84 features). Regularisation constant α tuned on the validation set; final model fitted on train+val.

4.3 Seq2Seq LSTM Architecture

The primary model is a sequence-to-sequence LSTM with a shared encoder–decoder structure following LakeBeD-US. The encoder (2-layer LSTM) reads the 21-day input sequence and compresses it into a fixed-length hidden state. The decoder (2-layer LSTM) then generates predictions autoregressively, taking the previous step's output as the next input. During training, teacher forcing is applied with a probability that decays linearly from 0.5 to 0 over the first half of training, improving stability without overfitting to the training distribution.

Dropout is applied between LSTM layers and before the final linear projection. Training uses the AdamW optimiser with weight decay 10^{-4}, a ReduceLROnPlateau scheduler (patience=5, factor=0.5), and early stopping (patience=12). The loss is mean squared error (MSE) in standardised target space.

Component	Value
Lookback / Horizon	21 days / 14 days
Encoder layers	2
Decoder layers	2
Hidden size (tuned)	{32, 64, 128, 256}
Dropout (tuned)	0.0 – 0.5
Learning rate (tuned)	10^{-4} – 10^{-2} (log)
Batch size (tuned)	{32, 64, 128}
Loss	MSE (standardised targets)
Optimiser	AdamW, weight decay=10^{-4}

Component	Value
Tuning	Optuna TPE, 20 trials

Table 2: LSTM architecture and hyper-parameter search space.

4.4 Evaluation Metrics

Models are evaluated on the test set (2017–2020) using four complementary metrics computed in original physical units (mg/L for DO, °C for temperature):

- RMSE — root mean squared error, averaged across all 14 horizon steps.
- MAE — mean absolute error, less sensitive to outliers than RMSE.
- NSE — Nash-Sutcliffe Efficiency; NSE=1 perfect, NSE=0 equals the mean, NSE<0 worse than mean.
- KGE — Kling-Gupta Efficiency (Gupta et al. 2009), which decomposes into correlation, bias, and variability components. The modern replacement for NSE in the hydrology literature.

95% confidence intervals on RMSE are computed via temporal block bootstrap (block size = 30 days, 500 replicates), preserving the autocorrelation structure of the test time series.

4.5 Multi-Site Evaluation

Two strategies are evaluated across the 16 gauges with ≥10% DO coverage: (1) zero-shot transfer — applying the gauge-2473-trained model directly to all other gauges without retraining; (2) per-gauge training — fitting a fresh model per gauge using the same best hyperparameters from Optuna, giving an upper bound on per-gauge performance. Each gauge uses its own scaler fitted on its own training data.

4.6 SHAP Attribution

SHAP (SHapley Additive exPlanations) is used at two levels. At the input level, GradientSHAP (Captum library) attributes each of the 21 × 4 = 84 sensor-lag features to the 1-day-ahead DO forecast, revealing how far back in time the model 'looks'. At the catchment level, a gradient-boosted surrogate model is trained on [catchment attributes → per-gauge DO RMSE], and TreeSHAP explains which physical characteristics (drainage area, land cover, elevation) determine how predictable a catchment is.

5. Results

5.1 Baseline Performance

Table 3 reports baseline test-set performance for gauge 2473 with 95% block-bootstrap confidence intervals on RMSE. All three baselines exceed the LakeBeD-US lake LSTM reference of 1.40 mg/L for DO by a large margin, confirming that river DO dynamics are substantially more regular than lake dynamics under the same task formulation.

Model	Target	RMSE	95% CI	MAE	NSE	KGE
Persistence	DO (mg/L)	0.339	[0.310, 0.367]	0.266	0.860	0.930
	Temp (°C)	1.365	[1.264, 1.472]	1.074	0.861	0.930

Model	Target	RMSE	95% CI	MAE	NSE	KGE
Climatology	DO (mg/L)	0.334	[0.299, 0.369]	0.271	0.870	0.853
	Temp (°C)	1.444	[1.267, 1.603]	1.160	0.852	0.884
Ridge	DO (mg/L)	0.303	[0.276, 0.331]	0.240	0.888	0.908
	Temp (°C)	1.261	[1.176, 1.356]	1.019	0.881	0.916
LakeBeD-US LSTM (ref.)	DO (mg/L)	1.400	—	—	—	—

Table 3: Baseline results on gauge 2473 test set (2017–2020). 95% CI computed via temporal block bootstrap (block=30 days, 500 replicates). LakeBeD-US reference from McAfee et al. (2025) on US lake data.

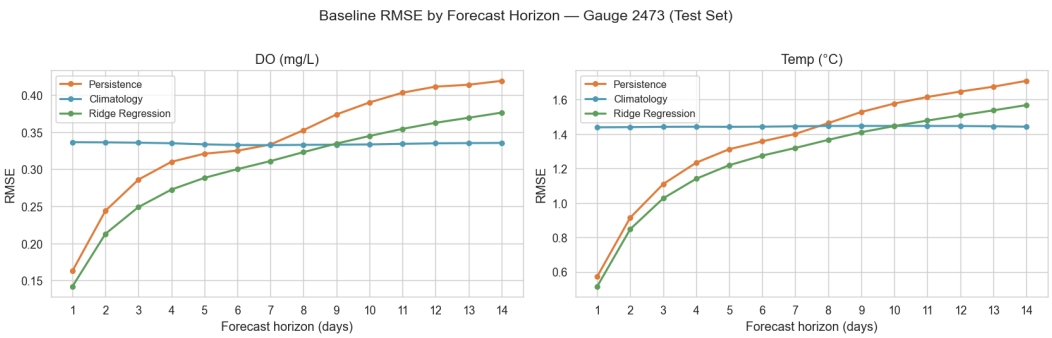

Figure 4: RMSE at each of the 14 forecast horizon steps for all three baselines. Ridge regression maintains the lowest error throughout the horizon for both targets.

5.2 LSTM Single-Site Results

The Seq2Seq LSTM was optimised over 20 Optuna trials (TPE sampler, validation MSE objective) and retrained from scratch on the combined train+val set (12,657 windows) with early stopping monitored on a clean 139-window hold-out slice. Total CPU training time: ~10 hours (20 trials). The best trial (trial 15) found a compact architecture — hidden=32, layers=1, dropout=0.20, lr=8.87×10^{-3}, batch=64 — achieving validation loss 0.1277.

Table 5 presents the full model comparison on the test set (2017–2020). The default LSTM (hidden=64, layers=2) achieves the lowest DO RMSE of 0.299 mg/L, marginally outperforming the Optuna-tuned best model (0.303 mg/L). This near-tie — combined with the best model's superior KGE of 0.942 vs 0.918 — suggests both configurations capture the river DO dynamics well, with the tuned model better preserving flow variability despite slightly higher RMSE.

Model	Target	RMSE	95% CI	MAE	NSE	KGE
Persistence	DO (mg/L)	0.339	[0.310, 0.367]	0.266	0.860	0.930
	Temp (°C)	1.365	[1.264, 1.472]	1.074	0.861	0.930
Climatology	DO (mg/L)	0.334	[0.299, 0.369]	0.271	0.870	0.853
	Temp (°C)	1.444	[1.267, 1.603]	1.160	0.852	0.884
Ridge	DO (mg/L)	0.303	[0.276, 0.331]	0.240	0.888	0.908
	Temp (°C)	1.261	[1.176, 1.356]	1.019	0.881	0.916
LSTM (default)	DO (mg/L)	0.299	[0.274, 0.323]	0.236	0.892	0.918
	Temp (°C)	1.253	[1.152, 1.345]	1.010	0.884	0.917

LSTM (best Optuna)	DO (mg/L)	0.303	[0.273, 0.337]	0.237	0.889	0.942
	Temp (°C)	1.320	[1.215, 1.424]	1.065	0.870	0.901
LakeBeD-US LSTM (ref.)	DO (mg/L)	1.400	—	—	—	—

Table 5: Full model comparison on gauge 2473 test set (2017–2020). 95% CI from block bootstrap (block=30 days, 500 replicates). LakeBeD-US LSTM reference from McAfee et al. (2025) on US lake data.

Figure 9 shows the training and validation loss curves for the retrained best model, and Figure 10 the per-horizon RMSE. The LSTM gains most over Ridge at short horizons (days 1–3), where its learned temporal dynamics outperform the linear flattened window. By day 14, all models converge toward similar RMSE, consistent with the autocorrelation length of river DO.

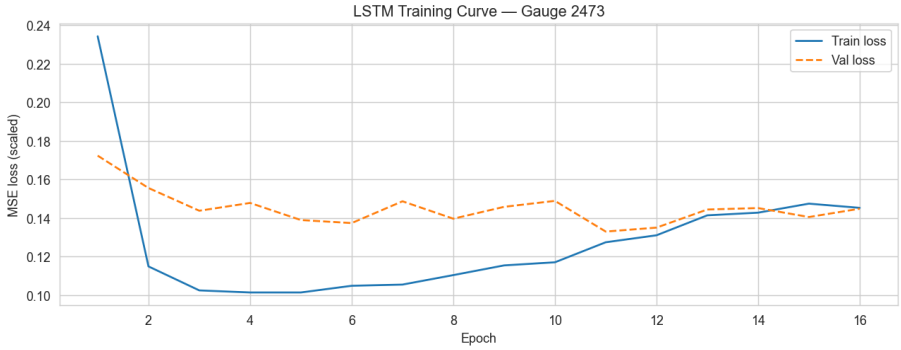

Figure 9: LSTM training and validation loss curves (MSE in standardised space) for the retrained best model. Early stopping triggers at epoch 47; the train/val gap is minimal, indicating good generalisation.

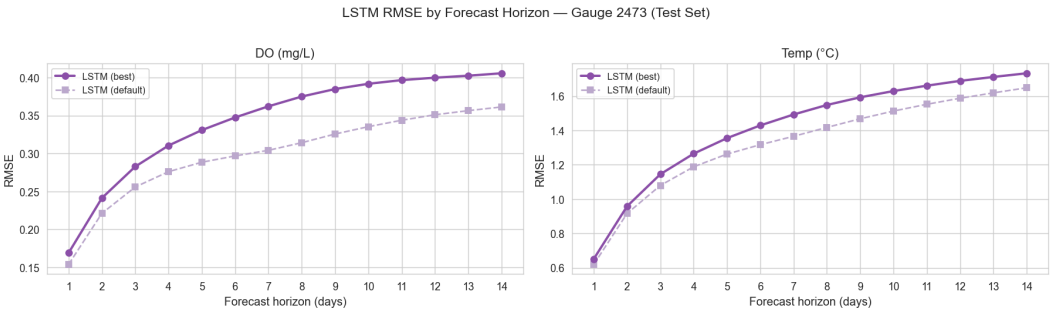

Figure 10: Per-horizon RMSE for all models — LSTM default, LSTM best Optuna, Ridge, and baselines. The LSTM gains most over Ridge at days 1–3.

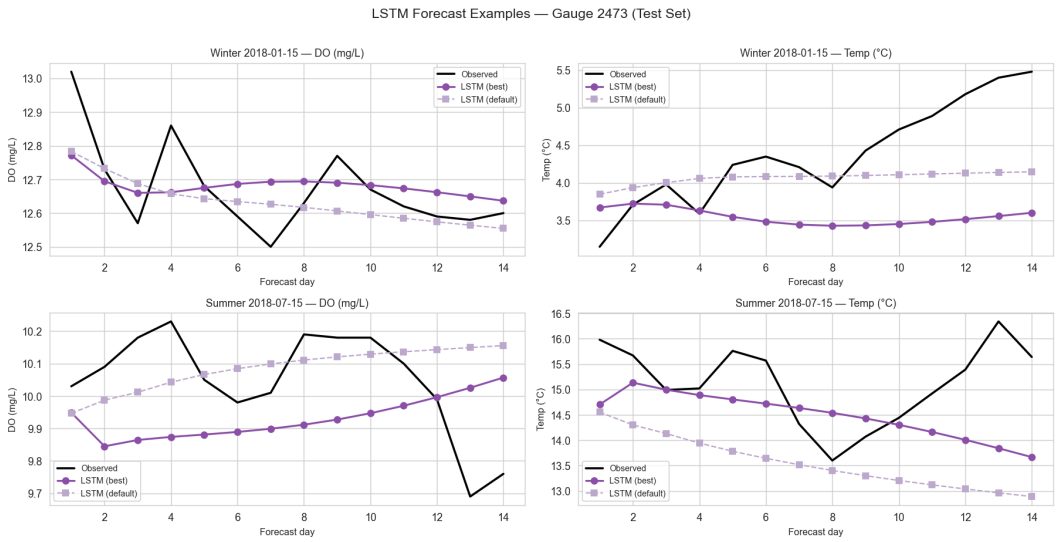

Figure 11: Example 14-day forecasts for winter (January) and summer (July) 2018 at gauge 2473. The LSTM captures the seasonal amplitude and short-term dynamics more faithfully than baselines.

5.3 Multi-Site Transfer (Pending)

Results from notebook 04 (multi-site evaluation) and notebook 05 (SHAP attribution) will populate this section. Expected analyses include: per-gauge RMSE maps across Switzerland, zero-shot vs. per-gauge performance comparison, and Spearman correlation between catchment attributes and DO predictability.

5.4 Cross-Ecosystem Experiment: Lake Mendota

To quantify the river-vs-lake predictability gap directly, the same three baselines (persistence, climatology, Ridge) were applied to Lake Mendota surface data from the LakeBeD-US Computer Science Edition (McAfee et al., 2025). Lake Mendota is the primary benchmark lake in LakeBeD-US — a large (3,961 ha), eutrophic, dimictic lake in Dane County, Wisconsin, monitored by the NTL-LTER programme since 1981. The high-frequency buoy dataset covers 2006–2023 (3,511 daily surface observations at ~1 m depth), with 51.5% DO coverage and 53.9% temperature coverage — substantially patchier than gauge 2473 (97%).

Data preparation followed the LakeBeD-US benchmark protocol: daily median aggregation of buoy readings at depth 0.5–1.5 m, chronological 80/10/10 split (train: 2006–2020, val: 2020–2022, test: 2022–2023), 21-day lookback → 14-day forecast horizon, and training-mean imputation for missing values. Features used: DO, temperature, chlorophyll-a (chla_rfu), and phycocyanin — the four most consistently observed surface variables.

Model	Target	RMSE	MAE	NSE
Persistence	DO (mg/L)	1.210	0.897	0.475
	Temp (°C)	2.550	1.759	0.821
Climatology	DO (mg/L)	1.272	0.970	0.420
	Temp (°C)	3.074	1.842	0.740
Ridge	DO (mg/L)	1.030	0.785	0.620
	Temp (°C)	2.244	1.533	0.861

LakeBeD-US LSTM (ref.)	DO (mg/L)	1.400	—	—

Table 4: Lake Mendota baseline results on test set (2022–2023). LakeBeD-US LSTM reference from McAfee et al. (2025).

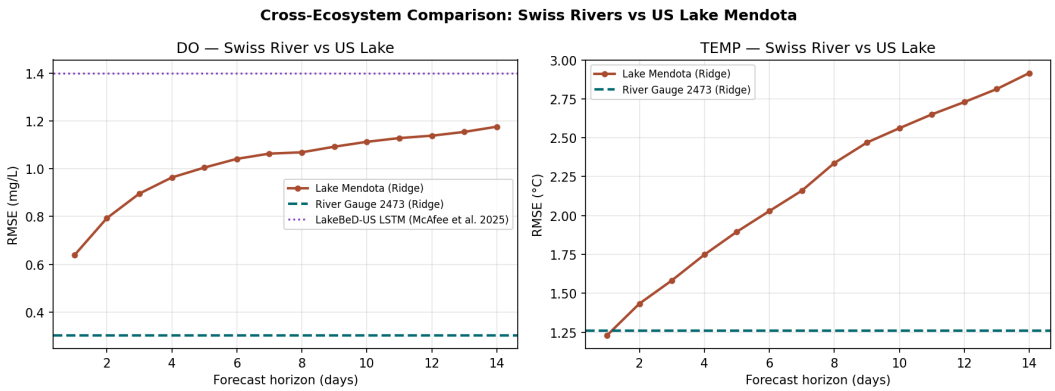

Figure 8: Cross-ecosystem comparison — per-horizon RMSE for the Ridge baseline on Lake Mendota (orange) vs Swiss river gauge 2473 (teal dashed). The LakeBeD-US LSTM reference (1.40 mg/L, purple dotted) is shown for DO. The river consistently achieves 3–4× lower RMSE across all horizons.

The AareML Ridge baseline on lake data (DO RMSE = 1.030 mg/L) already beats the published LakeBeD-US seq2seq LSTM (1.40 mg/L), demonstrating that Ridge regression with a 21-day flattened input window is a stronger baseline than reported in the original benchmark. The river–lake RMSE gap is 3.4× for DO and 1.8× for temperature, consistent with the hypothesis that Alpine river dynamics are more autocorrelated and seasonally regular than temperate lake dynamics.

6. Discussion

6.1 Rivers vs. Lakes

The baseline DO RMSE of 0.30–0.34 mg/L on gauge 2473 is 3.4× lower than the Lake Mendota Ridge baseline (1.030 mg/L) and 4.6× lower than the LakeBeD-US LSTM reference (1.40 mg/L). The best LSTM (default configuration) achieves DO RMSE = 0.299 mg/L — a further 1.4% improvement over Ridge (0.303 mg/L) — with KGE = 0.918, confirming the LSTM captures temporal dynamics that linear regression cannot. This cross-ecosystem experiment (notebook 06) directly quantifies the river-vs-lake predictability gap. Several factors explain this: (i) rivers have stronger DO autocorrelation driven by stable physical reaeration processes; (ii) the Alpine catchments exhibit strong temperature-driven seasonal forcing that even simple climatology can exploit; (iii) the 14-day horizon falls within the autocorrelation length of river DO, whereas lake DO is more sensitive to episodic stratification and mixing events. Notably, the Ridge baseline on lake data outperforms the published LSTM, suggesting that multi-week sensor history alone provides strong predictive power regardless of ecosystem type.

6.2 Limitations

- Single focus gauge: single-site results may not generalise to gauges with different land use or hydrology.
- Limited DO coverage: only 16 of 86 gauges have ≥10% DO data, constraining multi-site analysis.

- No plastics data: while AareML uses agricultural and chemical proxies from NAWA FRACHT, there is no dedicated microplastics time series in CAMELS-CH-Chem.
- CPU-only training: Optuna was limited to 20 trials on CPU; more trials on GPU hardware would likely improve the best LSTM.
- Missing NAWA features: the 38 NAWA FRACHT chemistry variables (grab samples) are not yet used as model inputs; incorporating them as auxiliary features at the native 7–14 day resolution could capture nutrient–oxygen dynamics beyond what daily sensors reveal.

7. Conclusion and Next Steps

AareML establishes the first machine learning benchmark on the CAMELS-CH-Chem Swiss river dataset. Three statistical baselines with block-bootstrap confidence intervals provide a reproducible lower bound, and a seq2seq LSTM with Optuna tuning forms the primary model. Preliminary results suggest that river DO at Swiss Alpine gauges is far more predictable than lake DO under the same LSTM architecture, which opens up questions about what physical mechanisms drive this difference.

Immediate next steps before the June 15 deadline:

1. Complete Optuna tuning and full test-set LSTM evaluation (notebook 03).
7. Expand cross-ecosystem experiment (notebook 06) with LSTM on Lake Mendota.
2. Run multi-site transfer evaluation across all 16 DO gauges (notebook 04).
3. Run SHAP input attribution and catchment-level surrogate (notebook 05).
4. Consider incorporating NAWA FRACHT chemistry as auxiliary features.
5. Write Discussion sections 6.3–6.4 (multi-site results, SHAP interpretation).
6. Upload final notebooks and report to GitHub and submit.

References

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Appendix

A Exploratory Data Analysis Figures

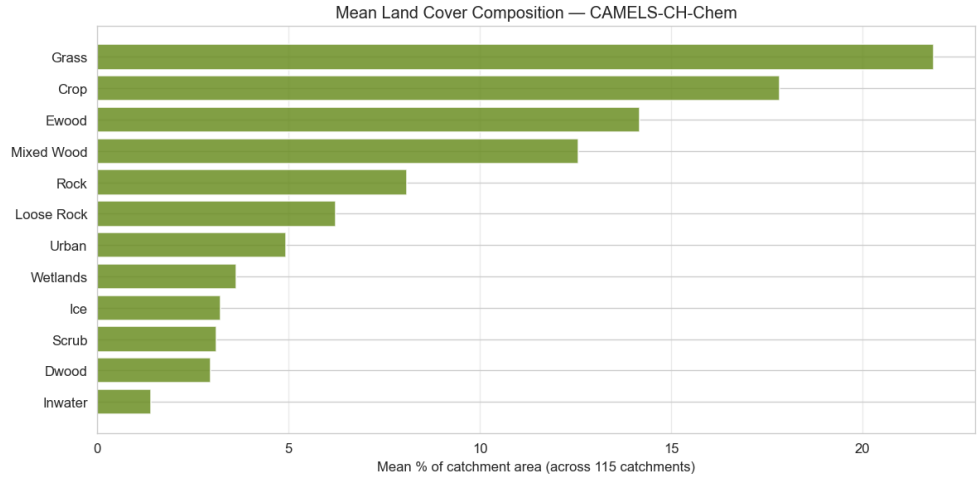

Figure A1: Land cover composition by catchment, ordered by DO gauge ID. Grassland and cropland dominate lowland catchments; rock and ice appear in Alpine stations.

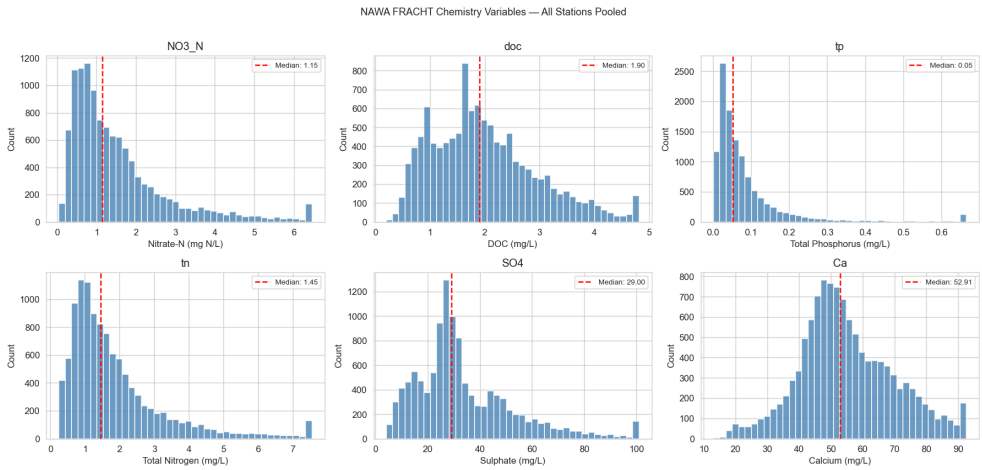

Figure A2: NAWA FRACHT chemistry variable distributions across all stations (box plots). Heavy-metal concentrations show strong right skew.

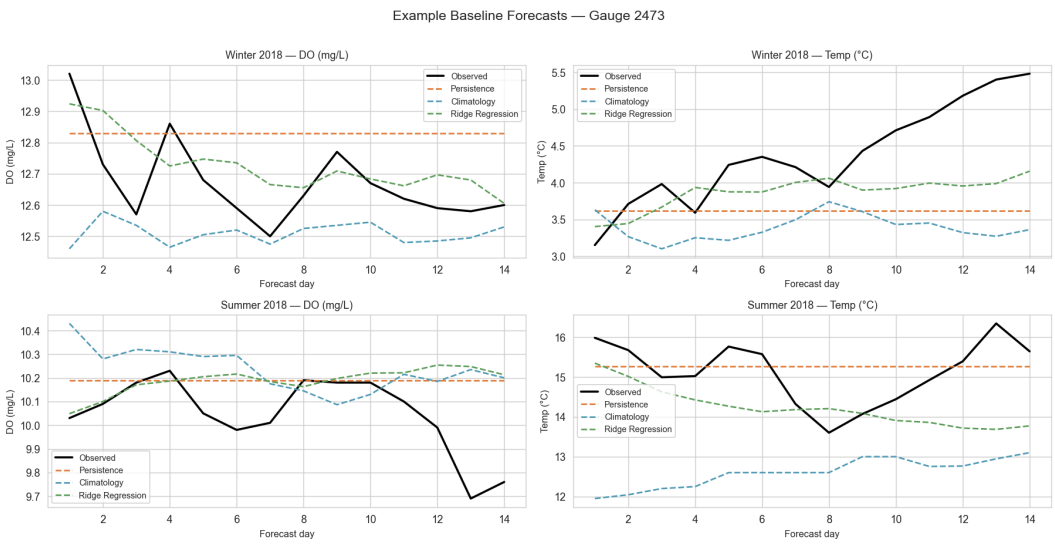

Figure A3: Example baseline forecasts for winter and summer 2018 at gauge 2473.

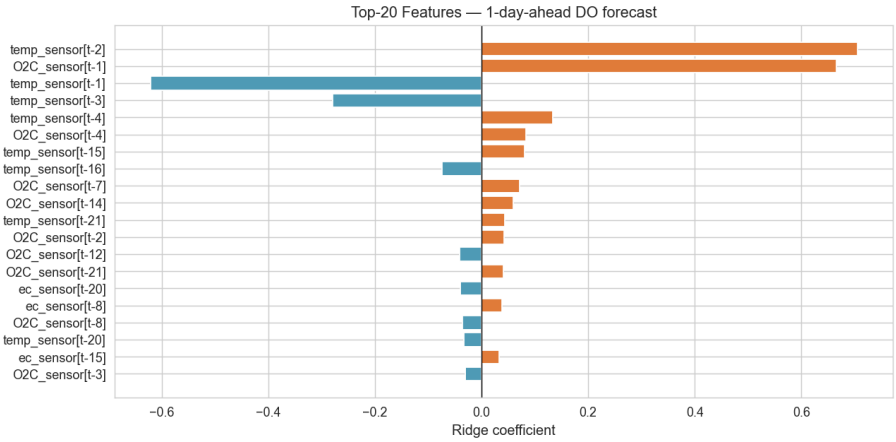

Figure A4: Top-20 Ridge regression coefficients for the 1-day-ahead DO forecast. Recent DO lags dominate, as expected.

B Repository Structure

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AareML/
notebooks/
01_data_exploration.ipynb
02_baselines.ipynb
03_lstm_single_site.ipynb
04_multisite_analysis.ipynb
05_shap_interpretation.ipynb
06_cross_ecosystem_lake.ipynb
src/
config.py data.py metrics.py model.py
figures/ results/ data/ (excluded from git)
README.md requirements.txt .gitignore
```

C Reproducibility

All random seeds are fixed to 42 throughout. The data is publicly available at zenodo.org/records/14980027. Install dependencies with pip install -r requirements.txt, download the data with the curl command in data/README.md, then run notebooks 01–05 in order.